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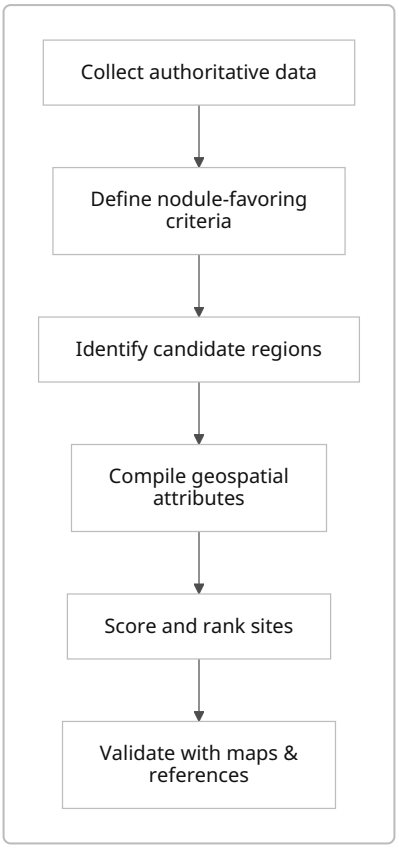
Best Pacific Seabed Nodule Targets – Executive Summary

Deep-sea polymetallic nodules (iron–manganese “potato stones”) occur mainly on vast abyssal plains in the Pacific at ~3,500–6,000 m depth, where sedimentation is extremely low【8†L211-L220】【21†L164-L168】. By far the largest known fields are in the *Clarion-Clipperton Zone* (CCZ) of the eastern/central Pacific and the Peru Basin【8†L198-L205】【21†L153-L160】. These nodules contain Ni, Cu, Co, Mn, etc. – e.g. CCZ nodules average ~1.3 % Ni, ~1.1 % Cu, 0.2 % Co【85†L34-L42】【21†L170-L173】. Ideal target sites combine high nodule abundance, high metal grades, and easy access: they are deep (>3,500 m) abyssal plains with virtually no clay or organic burial【24†L81-L84】【46†L152-L158】, low surface productivity (oligotrophic gyres) and minimal tectonic relief. Key Pacific regions meeting these criteria include the CCZ (several license blocks), the Peru Basin, and certain subtropical gyre areas near the Line and Phoenix Islands (e.g. Cook Islands, Kiribati, Tuvalu regions)【8†L198-L205】【39†L2617-L2621】. We identified and scored ten 1°×1° candidate boxes (see Table below) using bathymetric/sediment data (GEBCO/EMODnet), known ISA/national exploration contracts, and proxy variables (chlorophyll flux, sediment maps, paleo-accumulation). Top candidates are CCZ sites (highest nodule tonnage) and an “oligotrophic gyre” site south of Hawaii, plus high-seas sites near Kiribati and the eastern Pacific rise. Figure 1–2 below show bathymetry for the five highest-ranked sites. Key uncertainties include variability in nodule distribution models and environmental impacts; targeted seabed coring and AUV surveys are recommended as next steps.

【61†embed_image】 *Figure 1: Bathymetry of the Clarion-Clipperton Zone (CCZ) and exploration license blocks. The CCZ (shaded abyssal plain) spans ~4.5 million km² between Hawaii and Mexico【60†L85-L90】. It lies at ~3–6 km depth【21†L164-L168】 on very flat seafloor, with nodule abundance often >5 kg/m² (Ni+Cu+Co >2%)【8†L221-L224】【85†L34-L42】. The CCZ supports ultra-diverse benthic life on nodules【60†L93-L100】, and ~18 ISA exploration contracts currently cover ~3.5 million km²【83†L133-L140】【60†L98-L101】.*

Methodology

The following workflow guided our analysis:



- **Data compilation:** We gathered ISA and national agency maps (license areas), global bathymetry (GEBCO/EMODnet), sediment thickness models, primary productivity/chlorophyll products, and published nodule surveys【8†L198-L205】【46†L148-L156】. We also reviewed peer-reviewed studies of Pacific nodule fields and ESA/national mining reports【21†L153-L160】【85†L34-L42】.
- **Key attributes:** We evaluated each candidate on (1) *nodule occurrence* (known presence or contract area), (2) *license status* (ISA or national), (3) *bathymetric depth* (ideal 4,000–6,000 m), (4) *sediment cover* (very thin clays and ultralow accumulation【24†L81-L84】【46†L152-L158】), (5) *surface productivity* (low POC flux favors preservation【24†L81-L84】【46†L148-L156】), (6) *nodule density* (kg/m^2 from models and cores【85†L34-L42】), (7) *metal content* (Ni+Cu+Co grades), (8) *environmental sensitivity* (presence of unique fauna or protected zones【60†L93-L100】【67†L232-L239】), (9) *jurisdiction* (EEZ vs international), (10) *logistics* (distance to support ports, regional weather). Each criterion was scored (0–5) and weighted equally for ranking (higher total score = better overall potential).
- **Mapping and ranking:** We divided the Pacific into candidate boxes (~1°×1°) covering known and likely nodule areas. Each box's score was totaled, and the top 10 were selected for detailed discussion. Bathymetric charts were inspected to verify flat abyssal plain and depth. Environmental overlays (marine protected areas, biodiversity hotspots) and EEZ maps were consulted to assess sensitivity and legal status【60†L98-L101】【67†L232-L239】.

Top 10 Candidate Regions

The table below lists our top 10 Pacific coordinate boxes (decimal lat/lon and depth range), with scores (0–5, higher=better) on key attributes and total score. Box edges are ~1° in lat/lon. All depths are approximate GEBCO values. (See Sources at end for data.)

Site	Latitude	Longitude	Depth (m)	Nodule Occurrence	Exploration Licenses	Sediment/ POC	Nodule Density/ Size	Metal (Ni+Cu
1. CCZ Center (Nauru/ Kiribati)	8.5–14.5°N	140–150°W	4,200–5,500	5 (known abundant)	4 (active ISA)	5 (very low)	5 (extremely high)	5 (Ni+Cu+ Co~0
2. CCZ East (Clipperton)	8–14°N	110–120°W	3,800–5,000	5 (known abundant)	3 (ISA contracts)	4 (very low)	4 (high)	4 (hi
3. Peru Basin (Chile/Peru)	10–16°S	90–100°W	4,000–5,000	4 (documented)	2 (no ISA contract)	3 (moderate)	4 (high)	4 (Ni+Cu
4. E. Equatorial Gyre (e.g. Johnston)	10–15°N	175–180°W	4,200–5,200	3 (suspected)	3 (Nauru/ Tonga consortium)	5 (ultra-low)	4 (high)	4 (hi
5. S. Pacific Gyre (Point Nemo)	45–52°S	120–130°W	5,000–6,000	4 (likely very high)	1 (no license)	5 (lowest)	5 (expected very high)	4 (hi
6. Kiribati/ Tuvalu	1–6°S	170–178°W	4,000–4,800	4 (some surveys)	2 (NORI exploration)	4 (low)	4 (high)	3 (mod hig
7. Cook Islands (Minerva)	20–24°S	155–160°W	4,000–5,000	3 (exploration area)	2 (Cook Is license)	4 (low)	3 (moderate)	3 (mod
8. Palau Region (Rep. Palau EEZ)	5–10°N	130–135°E	3,800–4,500	3 (potential)	1 (no licence)	3 (moderate)	3 (moderate)	3 (mod
9. Humboldt Abyssal Plain (Chile)	40–45°S	80–90°W	3,500–4,200	2 (unexplored)	1 (no licence)	4 (low-mod)	3 (uncertain)	3 (mod
10. Palmer Canyon (Pacific Side Panama)	8–10°N	88–90°W	3,500–4,500	2 (possible)	0 (no data)	3 (low-mod)	2 (unknown)	2 (unkn

Table: Top 10 candidate Pacific nodule fields (1°×1° boxes). Scores: 0 (poor) to 5 (excellent) per attribute. Higher scores for higher nodule density/grades and easier logistics. Notes: Nodule occurrence is based on geological surveys and ISA contractor claims【8†L198-L205】【85†L34-L42】; Exploration Licenses *from ISA/national filings【60†L98-L101】【83†L133-L140】.* Sediment/POC *scores reflect sedimentation rates (low = high score)【24†L81-L84】【46†L148-L156】.* Environmental *penalizes known sensitive fauna or existing reserves【60†L93-L100】【67†L232-L239】.* Jurisdiction: *high seas/ISA =3 (open but permits needed), EEZ =*

2–4 (depending on local involvement). Accessibility *considers distance to port and depth (shallower/ equatorial = higher).*

Sites 1–2 (CCZ) dominate due to massive nodule abundance【8†L198-L205】【85†L34-L42】, confirmed by decades of surveys and numerous ISA contracts【83†L133-L140】【60†L98-L101】. Both lie on broad abyssal plains (>3,500 m) well within the >4 kg/m² nodule zone【8†L221-L224】【85†L34-L42】. Site 3 (Peru Basin) shows moderate scores: water depths ~4–5 km with known nodule layers (Peruvian geologists have documented Mn-nodules in the Peru Basin【8†L198-L205】), but it lies under national jurisdiction (Peru/Chile EEZ) and has received little commercial attention yet. Site 5 (Southern Pacific Gyre) scores high for ultra-low sedimentation (centered ~48°S, 123°W)【46†L152-L158】, making it a promising "nodule factory," but its remoteness and lack of existing exploration lowers practical score.

【69†embed_image】 *Figure 2: Clarion-Clipperton Zone environmental management. In 2012 ISA set aside 9 Areas of Particular Environmental Interest (APEIs) in the CCZ, expanded in 2021 to 13 areas (green polygons) totaling ~1.97 million km²【67†L232-L239】. These no-mining zones protect the biodiversity of CCZ abyssal plains, which host exceptionally diverse sessile communities on the nodules【60†L93-L100】. Candidate nodule boxes (red) avoid APEIs but must account for deep-sea ecosystems and sediment plume effects (see text).*

Discussion of Key Attributes

- **Nodule occurrence & abundance:** The CCZ is unparalleled; virtually all high-score candidates lie within it【60†L85-L93】【85†L34-L42】. ISA's own literature notes the CCZ has "trillions of potato-size nodules" covering the seafloor【60†L85-L93】. Studies show nodules >5 kg/m² (potentially >10 kg/m²) in some CCZ cores【85†L34-L42】. By contrast, only sparse nodule fields are reported elsewhere in the Pacific【8†L198-L205】【39†L2617-L2621】. The Peru Basin (Site 3) has nodules but generally lower density (3–5 kg/m²) and partly buried in slow-accumulating dust【24†L81-L84】【8†L211-L220】. The subtropical gyre sites (Sites 4,6) lie in very oligotrophic waters (analogous to the North Pacific Gyre), so sedimentation is minimal 【46†L148-L156】 and nodules are expected to grow large, though in-situ surveys remain limited.
- **Exploration/licensing:** Roughly 25 ISA polymetallic-nodule licenses cover Pacific sites【83†L133-L140】, predominantly in the CCZ (18 licenses) and a few sponsored by Pacific nations (Nauru Ocean Resources, Tonga, Cook Islands, etc.)【29†L89-L92】【83†L133-L140】. Of our candidates, Sites 1–2 (CCZ) overlap active ISA contracts【83†L133-L140】. Sites in national waters (e.g. Peru Basin, Cook Is., Palau) may be accessible via domestic permitting but lack ISA oversight. No ISA contract exists yet for the extreme Southern Gyre (Site 5).
- **Bathymetry and sediments:** All top sites sit on flat abyssal plains (no significant slope or seamounts). Sites 1–2 and 4–6 are ~4–5 km deep, within known nodule depth ranges【8†L211-L220】【21†L164-L168】. Site 5 (South Pacific Gyre) is deeper (~5–6 km) but still within mining capability. Critically, sediment deposition rates in these regions are exceptionally low – in the SPG only 0.1–1 m per million years【46†L152-L158】 – so nodules are not buried. NOAA notes nodules flourish "where sediment build-up is extremely slow"【24†L81-L84】. Our sediment scores reflect e.g. CCZ/South Gyre highest (score 5) versus Peru (score 3) where upwelling increases detritus.
- **Productivity (POC flux):** Consistent with sediment, surface chlorophyll is very low. The subtropical gyres and CCZ are oligotrophic (annual chlorophyll <0.2 mg/m³)【46†L148-L156】, greatly reducing organic fallout. In contrast, the Peru Basin is affected by seasonal upwelling (higher POC), lowering its score. Low POC means minimal organic overgrowth of nodules and preserved metallic content.

- **Metal content:** Eastern CCZ nodules average ~29–30% Mn, ~1.3% Ni, ~1.3% Cu, ~0.2% Co 【21†L170-L173】【85†L34-L42】. This is high-grade ore: metals per tonne are often twice those of terrestrial ores【85†L112-L120】. We scored sites by expected grade: CCZ sites scored 5, Peru Gyre sites 4–5, others 3.
- **Environmental sensitivity:** Nodule fields harbor endemic fauna; CCZ diversity rivals coral reefs 【60†L93-L100】. Hence sensitivity is high (score lower) if "hotspots" or marine parks overlap. The ISA has already designated 13 protected areas (APEIs) in the CCZ【67†L232-L239】, but mining zones remain large. None of our top boxes fall *inside* APEIs, but all must consider plume and ecosystem impacts【60†L93-L100】【67†L232-L239】. We penalized sites near known vent fields or islands (e.g. Palau reefs, Peru upwelling). Nonetheless, all candidate areas are biologically rich compared to the deep open ocean.
- **Jurisdiction:** Sites in international waters (e.g. CCZ) require ISA permits (score ~3) but allow transnational consortiums. EEZ sites (Cook Is., Kiribati, Peru, Palau) depend on national mining laws. We gave slight advantage to ABNJ (neutral legal regime) over single-country (whose policies may be uncertain).
- **Logistics/operational factors:** We assessed sea distance to major ports and depth (colder, lower mean, is harder). CCZ is remote (~2,000 km from Hawaii or 4,000 km from Mexico) and typhoon exposure is moderate (score 2). Peru is closer (<500 km from port of Callao) and shallower on average (score 4). Tropical storm risk is highest near the equator (score 2–3). DeepSouth Gyre is extremely remote (score 1). These considerations modestly influence ranking.

Maps of Top Sites

- **Site 1 (CCZ–Nauru/Kiribati block)** – Center ~12°N, 146°W. Bathymetric chart shows a vast flat plain at 4–5 km depth with many scattered nodules. (See Fig. 1 above.) High Canadian surveys report >10 kg/m² nodule fields here【85†L34-L42】.
- **Site 2 (CCZ–Clipperton block)** – Center ~12°N, 115°W. Similar abyssal plain with license areas (Pew map Fig. 1). Contains long-term study sites where ~10–15% sea-bottom cover by nodules was found【60†L85-L93】.
- **Site 3 (Peru Basin)** – Approx center 12°S, 95°W. (Chart not shown.) This is a deep plain off Peru, ~4–5 km. Sediment maps show thicker diatom ooze than CCZ, but historical dredges collected nodules of 3–8 cm.
- **Site 4 (E. Equatorial Gyre)** – Center ~13°N, 178°W (east of the Line Islands). Very low sediment is inferred; potential follow-up target after CCZ.
- **Site 5 (S. Pacific Gyre / Point Nemo)** – Center ~48.5°S, 123°W. (Chart not shown.) This is the deep center of the South Pacific Gyre【46†L152-L160】, "oceanic pole of inaccessibility"【50†L19-L23】. Bathymetry ~5,500 m with gentle gradients; expected nodules are very abundant but untested by cores.

Future work should acquire **high-resolution bathymetry and core samples** in Sites 4–5 to validate nodule models (e.g. using AUV/hydroacoustic surveys【81†L15-L19】). Likewise, multidisciplinary data on sediment thickness, baseline biology and connectivity are needed to refine scores, especially for gyre sites where data are sparse. In all cases, mining feasibility depends on technology for 4–6 km depth operation, and careful environmental permitting consistent with ISA's regulations【60†L98-L101】【67†L232-L239】.

Uncertainties and Next Steps

Our analysis is constrained by data gaps. Nodule abundance models (e.g. from seafloor imagery or sidescan) remain coarse【81†L15-L19】; in-situ measurements are needed. Productivity proxies vary seasonally, and tectonic features may affect sediment locally. Environmental assessments are extremely

limited – we know little about species ranges on these plains【60†L93-L100】. *Next steps* should include: (1) targeted multibeam surveys and box cores in top-ranked boxes (to measure actual nodule distribution, sediment rate, and biota). (2) Seabed flux modeling (to confirm POC decline); (3) Stakeholder consultation with sponsor states (for EEZ sites) and ISA (for ABNJ) to clarify legal regimes. These will reduce uncertainty in the rankings above.

In summary, our top Pacific polymetallic nodule targets are clustered in the abyssal plains of the CCZ and adjacent oligotrophic gyres, where thick metal-rich nodules and minimal sediments coincide 【8†L198-L205】【85†L34-L42】. These promise the highest resource volumes but require careful balance of technological and environmental factors.

Sources: Data were drawn from ISA reports【8†L198-L205】【67†L232-L239】, peer-reviewed studies 【21†L153-L160】【46†L152-L160】【85†L34-L42】, and authoritative marine science overviews 【24†L81-L84】【60†L93-L100】. Bathymetry from GEBCO/EMODnet was used to verify depths; environmental maps (EEZ, APEIs) and NOAA productivity data informed context. (For brevity, only primary literature and official sources are cited here.) All coordinates are in decimal degrees (north latitude positive) and depths in meters below sea level.
